Cascade User Manual

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A complete guide to uploading, cleaning, visualizing and exporting data in Cascade, with every button, limit and tip in one place.

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1. Getting started

You sign in once, then work through three pages in order: Upload, Clean Data and Visualize. Buttons appear in capital letters in the app; this manual writes them in normal case.

Sign up and sign in

1. Open Cascade and press **Log in** in the top bar. **Upload your first file** also takes you to sign-in.
2. New user: on the sign-in page press **Sign Up**, type a username, type a password twice (Password and Password confirmation), then press **Sign Up**. You are signed in at once and land on the Upload page.
3. Returning user: type your Username and Password, then press **Sign In**.
4. To leave, press **Sign Out** at the top of the Upload page.

The sign-up form follows your site's password rules, which usually mean at least 8 characters, not all digits and not a common password.

Light and dark mode

The sun button (☼) at the top right of each page switches between light and dark. Cascade remembers your choice in that browser.

What your file must look like

Item	What Cascade expects
File types	.csv, .xlsx, .xls, .json, .jsonl, .ndjson
Size	5 MB at most. A bigger file is refused before anything is saved.
Layout	One table: column names in the first row, one record per row, one value per cell
Excel	Only the first sheet is read. Remove titles, merged cells and total rows around the table.
CSV	Comma-separated text in UTF-8
JSON	A list of records, JSON Lines, or an object that wraps a list. Nested objects become columns named parent_child.
Dates	Written dates are rewritten as YYYY-MM-DD. Excel date numbers such as 45292 are converted when the column name says Date.
Numbers	Currency symbols and thousands commas are removed and the currency symbol is remembered for display

Keep in mind

- Your charts, pivots, filters and notes live in your current sign-in. Signing out, clearing cookies or switching browser loses them. Use **Save Progress** (section 8) to keep them in a file.
- Uploaded and cleaned files are stored on the server that runs Cascade. Upload only data you are allowed to store there.
- Cascade lists every automatic change it makes and never fills in missing values for you.

2. Step 1: Upload

The Upload page (titled Data Importer) is where every session starts. It has two boxes: one for a new data file and one for saved work.

Upload a new file

1. Under **Select CSV, Excel, or JSON file to analyze**, choose your file.
2. Press **Upload & Clean Data**. Cascade checks the file and opens the Clean Data page.

Cascade shows the 5 MB limit under the file box and tests your choice the moment you pick it. If the file is empty or too large, the button turns off and a message appears so you can choose another file.

Continue saved work

Under **Select a Cascade progress file (.cascade)**, choose a file made earlier with Save Progress and press **Import & Continue**. You go straight to Visualize with your charts, pivots, filters and notes restored. Progress files can be up to 15 MB. See section 8.

Messages you may see

Message	Cause and fix
Please choose a file to upload.	Nothing was selected. Choose a file first.
That file is empty.	The file has no content. Check you exported the right one.
“name” is X MB, which is over the 5 MB limit.	Remove unused rows or columns, or split the data into parts.
Unsupported file type.	Save the file as .csv, .xlsx, .xls or .json.
That file is not valid JSON. / Could not turn that JSON into a table.	Use a list of records such as [{"region": "North", "sales": 100}].

Keep in mind

- Uploading a new file replaces your current work. Charts, pivots, filters, Key Metrics and edited notes from the previous file are cleared, so press **Save Progress** on Visualize first if you want to return to them.
- The upload only reads the file. Your original is never changed.

3. Step 2: Clean data

Press **Clean Data** and Cascade tidies the file automatically, then asks what to do with any rows that have missing values. Nothing is filled in for you, and nothing is applied until you confirm.

Buttons and links on this page

Button or link	What it does
Clean Data	Starts cleaning the file you uploaded
Upload a different file	Returns to Upload so you can choose another file
Cancel	On the missing-values screen: leaves without applying anything
Review & confirm	Checks your choice and opens the confirmation box
Go back / Confirm & apply	In the confirmation box: return and change your mind, or apply and save
Download Cleaned CSV	Saves the cleaned data as cascade_cleaned_data.csv
Proceed to Visualize →	Opens the Visualize page

What is fixed automatically

Step	What Cascade does
Empty rows	Removes rows where every cell is blank
Duplicates	Removes rows that are exact copies of another row
Spacing	Trims text and collapses repeated spaces
Blank markers	Treats blank, NA, N/A, na, n/a, null, NULL and "-" as missing
Numbers	Turns text that looks like numbers into numbers when 80% or more of a column's values qualify. Currency symbols and commas are removed, and the symbol is remembered.
Column names	Lowercases them and swaps spaces for underscores, so Order Date becomes order_date
Capitalization	In category-like columns, spellings that differ only by case are merged into the most common one
Dates	Excel date numbers in a column named like Date are converted. Date columns are rewritten as YYYY-MM-DD, or YYYY-MM when only a month is given.
Whole numbers	A column that is entirely whole numbers shows 21, not 21.0

Two things are flagged but never changed: possible outliers in numeric columns, and pairs of near-duplicate rows (most columns identical). Near-duplicates are only checked in files of up to 1,000 rows.

Missing values: your decision

If no row has a gap, cleaning finishes at once and shows the report. Otherwise a **Missing values found** screen shows how many rows, complete rows, missing cells and columns are affected, plus a table, **Where the gaps are**, with the missing count and share of rows for each column. Under **How do you want to handle these rows?** pick one option:

- **Remove N rows with missing values**: deletes every row that has a gap and keeps the complete rows.
- **Review and edit manually**: lets you go through the affected rows one by one.

Press **Review & confirm**. A box titled **Confirm your choices** states exactly what will happen, for example "Remove all 9 rows that have missing values. The cleaned data will have 511 rows." Press **Confirm & apply** to go ahead, or **Go back** to change it.

Reviewing rows manually

- For each row, choose **Edit** and type a value into every blank cell, or choose **Remove row**.
- Move around with ← **Previous**, **Next** → and **Next unfinished row**.
- Typed values are checked against the column: a whole number, a number, a date as YYYY-MM-DD, a month as YYYY-MM, or text. Text is matched to the capitalization already used, so "south" becomes "South".
- A value that would be read as missing, such as NA, null or a dash, is refused.
- Every row needs a decision before **Review & confirm** continues.
- Row numbers match your source file, so Row 33 is row 33 in your spreadsheet.
- Manual review handles up to 500 rows at a time. With more, remove the rows or fix the source file and upload it again.

The cleaning report

After you confirm, the **Cleaning Report** shows original and cleaned rows and columns, **Your Missing-Value Decisions**, and a list under **What Cascade Cleaned**. Visualize, the PDF and Download Cleaned CSV all use this cleaned data, and the same notes appear in File Overview and in the PDF.

Keep in mind

- Check the removed-duplicate and removed-row counts. Two genuine orders that are identical in every column look like duplicates, so give every record an ID column in your source file.
- Column names change. Use the new lowercase names when you build charts and pivots.
- Outlier flags are a prompt to look, not a verdict. Real large sales are outliers too.
- To start over with the same file, press **Clean Data** again. To use a different file, press **Upload a different file**.

4. Step 3: The Visualize page

Visualize is where the report is built. Everything on this page, the Key Metrics, charts, pivot tables, notes and recommendations, is what the PDF will contain.

Page layout

Area	What is in it
Top bar	The sun button (theme), **Download PDF**, **Save Progress**, **Import Progress** and **Upload New File**
Left panel (Contents)	**+ Add Chart** and **+ Add Pivot Table** at the top, then links to File Overview, Overall Summary, Key Metrics, Written Summary and Filters. Below them is the numbered **Charts & Pivots** list, then Recommendations, **Save Progress** and **Download PDF**.
Main column, in order	File Overview, Overall Summary (Key Metrics and Written Summary), Filters, your pivot tables and charts in the order you chose, Recommendations, then **+ Add Another Chart** and **+ Add Pivot Table**
Small screens	The left panel folds away. Press ☰ **Contents** to open it, or use the **+ Chart** and **+ Pivot** shortcuts.

Click any entry in the Contents panel to jump to that part of the page.

File Overview

Shows the Author (your username), file name, total rows and columns, the detected column names and a note on how you handled missing values. Press **Show Data Preview (First 10 Rows)** to open a small table of your data and press it again to hide it.

Key Metrics

Key Metrics are the headline number cards at the top of the Overall Summary.

- **Starting set:** the record count, the totals of the first three numeric columns, and a distinct count for the first suitable category column.
- **Add a card:** pick a column under **Add a card for**, pick a calculation, then press **Add Card**. Choose **Number of records** for a simple row count.
- **Calculations by column type:** a number column offers Total, Average, Median, Minimum, Maximum, Count (non-blank), Distinct count and Most common. A date column offers Minimum, Maximum, Count, Distinct count and Most common. A text column offers Count, Distinct count and Most common.
- **Remove a card** with the × in its corner. Press **Reset to auto** to return to the starting set.
- You can have up to 12 cards. A card that already exists is not added twice.
- Money columns show the currency symbol that was detected during cleaning, and large figures are shortened, for example ₹2.40M.
- When a filter is on, each card also shows the figure for all data and either its share of the whole or its change against the whole.

Written Summary

Cascade writes a short summary from your Key Metrics and charts. To use your own words, edit the box and press **Save Summary**. Press **Reset to Auto-Generated** to bring the automatic text back. The summary appears in the PDF.

Filters

Filters narrow the rows used by the Key Metrics, every chart, every pivot table and the PDF. A line above the controls reads, for example, "Showing 511 of 511 rows".

1. Under **Column**, choose the column to filter on. The boxes next to it change to suit the column.
2. For a number column type a **Min** and/or **Max**. For a date column choose **From** and/or **To**. For a text column tick the values to keep (**Select all** and **None** help), or, when the column has more than 200 different values, type text into **Value contains**.
3. Press **Add Filter**. It appears as a tag above the controls.

- Filters combine, so a row must pass all of them.
- A column can have only one filter. Adding another replaces it. The page keeps up to 12 filters.
- Remove a single filter with the × on its tag, or press **Clear all**.
- If the filters match no rows, charts show a message asking you to adjust or clear them, and the PDF cannot be produced until at least one row matches.

Keep in mind

- The starting Key Metrics are just the first numeric columns, so they may not be the numbers you care about. Remove those that mean nothing, such as a total of unit prices, and add the ones that matter.
- Settings change the notes. Adding or removing a card discards a saved Written Summary, and changing a filter discards every saved note. Cascade asks you to confirm before it resets anything you wrote, but it is easier to finish your Key Metrics and filters first and write your notes last (section 7).

5. Pivot tables

A pivot table adds up your data by one or two categories, such as revenue by region and sales channel. It always uses the rows left after your filters. A new session starts with Pivot Table #1, and you can add more.

Settings

Setting	What it controls
Rows	The column whose values become the table's rows. It can be any column.
Columns (optional)	A second column whose values become table columns. Leave it on (none) for a simple list.
Values	What is measured: **Number of records** or any column
Calculation	Sum, Average, Median, Minimum, Maximum, Count or Distinct count. Number columns offer all of these. Other columns offer Count and Distinct count.
Group dates by	Appears when Rows or Columns is a date column: No grouping, Daily, Weekly, Monthly, Quarterly or Yearly
Show	Show all, **Top N rows** or **Bottom N rows**. N can be 1 to 50.
Sort rows by	Total / value, or Row label
Order	High to low / Z to A, or Low to high / A to Z

Buttons

Button	What it does
Update Pivot	Applies the settings you changed. Nothing changes until you press it.
Reset	Returns this pivot to its starting settings and clears its saved notes
Download CSV	Saves the table exactly as shown, with filters applied, as cascade_pivot_table_N.csv
Duplicate Pivot	Copies the settings so you can build a companion table, for example the same table grouped differently
Remove Pivot	Deletes this pivot table
▲ and ▼	Move the pivot up or down among your charts and pivots
+ Add Pivot Table	Adds another pivot. It starts on a Rows column you have not used yet.

Reading the table

- The table lists each row with its **Rank**. Rank 1 is the highest value, and tied rows share a rank.
- A **% of total** column appears for Sum and Count, where the parts add up to a whole. It is left out for Average, Median, Minimum, Maximum and Distinct count.
- The last row is the **Grand total** (for Sum and Count) or **Overall** (for the others).
- With a Columns field, each column has its own value plus a Total column.
- A **Key insight** and **Recommendation** are written for each pivot. You can edit them (section 7).

Recipes

You want	Rows	Columns	Values	Calculation
Revenue by region	region	(none)	revenue	Sum
Revenue by region and channel	region	sales_channel	revenue	Sum
Average rating by category	product_category	(none)	customer_rating	Average
Orders per month	order_date, grouped Monthly	(none)	Number of records	Count
Five best products	product	(none)	revenue	Sum, Show Top N rows, N = 5

Limits

- With a Columns field, only the 15 columns with the largest totals are shown, and a note says how many were left out.
- With **Show all**, only the first 200 rows are shown, and a note says so. Use Top N, a filter or a coarser grouping to narrow it.
- Each pivot has one Rows field, one optional Columns field and one measure.

Keep in mind

- Changing a pivot's settings with **Update Pivot** replaces its saved insight and recommendation with fresh automatic text. Write your own after the settings are final.
- Averages, medians and extremes are not additive, so the table shows no share-of-total for them. Use Sum or Count when you want shares.
- Pivot tables are included in the PDF in the order you place them.

6. Charts

Each chart card has a settings strip, the chart itself and a Written Insights box. A new session starts with one chart. Charts are pictures drawn by Cascade, so they do not respond to hovering, and the PDF shows the same picture.

Working with charts

Button	What it does
+ Add Chart	Adds a chart. It starts on the first column against itself, so choose your own X and Y. It is in the left panel and at the bottom of the page.
Update Chart	Applies the settings you changed. Nothing changes until you press it.
Duplicate Chart	Copies the chart so you can try another chart type beside it
Remove Chart	Deletes the chart after you confirm. It shows only when there is more than one chart.
▲ and ▼	Move the chart up or down among your charts and pivots
⇄	Swaps the X and Y columns. A hint, "Looks reversed", with a **Switch axes?** link, appears when a numeric column is on X and a text column is on Y.

Chart types

Chart Type	Use it to	What you choose
Auto (Recommended)	Let Cascade pick from your columns. The card shows an "Auto:" badge naming its choice.	X and Y
Bar Chart	Compare totals across categories	X category, Y number. A text Y counts rows.
Horizontal Bar Chart	Compare many categories or long names	Same as Bar
Pie Chart	Show shares of a whole	**Categories** and **Values**
Line Chart	Follow a trend in order, usually over time	X date, Y number
Area Chart	A trend with the area under it filled	X date, Y number
Scatter Plot	See whether two numbers move together	X number, Y number
Histogram	See how one column's values are spread	**Column to Analyze** only
Box Plot	See spread, middle and outliers of a number, optionally per category	Y number, X category
Correlation Heatmap	See how strongly all numeric columns move together, from -1 to 1	Nothing. It needs at least two numeric columns.

When the type is Auto, Cascade chooses like this: the same number column on both axes gives a Histogram; a date on X and a number on Y gives a Line; two numbers give a Scatter; a category with a number gives a Bar, or a Horizontal Bar when the category has more than 12 values; anything else gives a Bar of counts.

Date grouping

When X is a date and Y is a number on an Auto, Line, Area or Bar chart, a **Date grouping** row appears. Choose **Group dates by** Daily, Weekly, Monthly, Quarterly or Yearly, then choose how to combine each period under **Show**: Sum (total), Average, Count of records, Maximum or Minimum. Without grouping, a line chart draws one point for every row, which is hard to read for long data.

Sort & limit

For Bar, Horizontal Bar and Pie charts, a **Sort & limit** row appears. **Sort by** offers Default order, Value high to low, Value low to high, Label A to Z and Label Z to A. **Show** offers Show all, Top N or Bottom N, with N from 1 to 50. A note on the chart says how many groups are shown, and the written insight refers only to those groups.

Without Sort & limit, a bar chart shows only the first 20 categories in label order, a horizontal bar chart the first 15, and a pie chart the 7 largest slices plus one called Other. Use Top N to see the biggest or smallest categories instead.

Calculated values (ƒx Calculate)

Use a calculation when the number you want is not a column, for example profit margin. The **ƒx Calculate** button sits under the Y Axis for Bar, Horizontal Bar, Pie, Line and Area charts, and under the X Axis for Scatter, Histogram and Box charts. It turns into "ƒx Calculating · turn off" while it is active.

1. Choose **Column A**, then an **Operation**: ÷ Divide, × Multiply, − Subtract or + Add.
2. Under **By**, choose **Column** or **Number** and pick the second column or type the number.
3. Tick **× 100 (show as %)** to show a percentage.
4. For a calculated Y, choose **Combine rows by**: **Totals** adds up each group first and then divides (a ratio of totals, which is right for a margin), while **Average of each row** works out every row and averages the results.
5. Press **Update Chart**.

Example: Profit ÷ Revenue × 100 by region, combined as Totals, gives each region's true profit margin.

- A calculated X is worked out for every row and then plotted, for example Unit Price × Quantity.
- Only one calculation can run at a time. If both are on, the Y one is kept.
- Dividing by zero is ignored, with a warning on the card.
- A pie cannot show ratios or averages, so Cascade draws it as a bar chart and says so.

Written Insights

Under every chart, Cascade writes a short insight from the data, in your currency symbol where one was detected. Edit the text and press **Save Chart & Insights**. A **Custom** tag shows next to text you have changed. The insight appears under the chart in the PDF.

Keep in mind

- Changing X, Y, type, grouping, combining, sort, limit or a calculation replaces the saved insight with new automatic text. Cascade asks "This will replace your saved/edited insight with a new auto-generated one. Continue?" first, but only when that chart already has a saved or typed insight. Set up the chart, then write your insight.
- Adding, duplicating or removing a chart resets a customised Overall Summary and Recommendations. Cascade asks first. Changing any chart's settings resets them too, and that one happens without a prompt, so settle your charts before you edit those two boxes.
- A pie of many categories groups the smaller ones into Other. Use Top N on a bar chart when you need to see every large item.
- Charts use the filtered rows, so a filter that is still on will change what every chart shows.

7. Arranging the page and editing notes

Charts and pivot tables share one order, and the PDF follows the same order as the page. Every written part of the report starts as automatic text that you can rewrite.

Arranging charts and pivot tables

1. Find your charts and pivots in the **Contents** list on the left, under **Charts & Pivots**. On a small screen, press ☰ **Contents** at the bottom to open it. Each entry shows its position number, its title and its type.
2. Drag an entry by its handle (⋮⋮) to a new place in the list. Or press ▲ or ▼ in the top corner of the chart or pivot card to move it one place. ▲ is greyed out on the first block and ▼ on the last.
3. Click any entry to jump straight to it.

The hint "Drag to rearrange, or use the ▲ ▼ buttons on each block" shows once you have two or more blocks.

- Charts and pivots can be mixed in any order. Only they move. File Overview, Overall Summary and Filters stay at the top and Recommendations stays at the bottom.
- Blocks are numbered by position, so "Chart 3" is whichever chart is third at the moment.
- Moving a block loses nothing. If your own recommendations start with "Chart 2 - …", Cascade renumbers that line to follow the chart.

Editing the written parts

Where	Box	Save button	Reset
Overall Summary	Written Summary	Save Summary	Reset to Auto-Generated
Each chart	Written Insights	Save Chart & Insights	Clear the box and save
Each pivot table	Key Insight and Recommendation	Save Pivot Insight & Recommendation	Reset to Auto-Generated
Recommendations	One recommendation per line	Save Recommendations	Reset to Auto-Generated

- Your own text shows a **Custom** tag. As soon as you type, the tag reads **Custom (unsaved)**, which is a reminder that nothing is stored until you press Save.
- **Reset to Auto-Generated** only appears once you have saved your own text.
- Saving an empty box returns it to the automatic text.
- In Recommendations, each line becomes one numbered item in the report and blank lines are dropped. A line that starts like the automatic ones, for example "Chart 2 - Bar Chart (sales by region): …", is also shown under that chart in the PDF. Any other line is labelled General.

What resets your notes

Automatic text describes the data as it is now, so Cascade throws away your edits when what they describe changes. Use this table to see which notes a change will touch.

If you…	These go back to automatic text	Asks first?
Add, duplicate or remove a chart	Overall Summary and Recommendations (a removed chart's own insight is deleted)	Yes, if either is custom
Change a chart's settings	That chart's insight, plus Overall Summary and Recommendations	Only for the chart's own insight
Add, change or remove a filter	Every chart insight, every pivot note, Overall Summary and Recommendations	Yes, if you have any custom text
Add or remove a Key Metric, or press Reset to auto	Written Summary	Yes, if it is custom
Change a pivot table's settings, or press Reset pivot	That pivot's Key Insight and Recommendation	No
Duplicate a pivot table	The copy starts with automatic notes	No

If you…	These go back to automatic text	Asks first?
Move a chart or pivot	Nothing	No
Add or remove a pivot table	Nothing (a removed pivot's notes are deleted)	Yes, when removing

Keep in mind

- Work in this order: filters, Key Metrics, charts and pivots first, then the written parts. Writing notes last means nothing is thrown away.
- Automatic text is a good first draft, so read it before you replace it. It quotes real numbers from your filtered data.
- Notes are kept in your session. They survive moving around the page, but they are lost when you upload a new file. Use **Save Progress** (section 8) to keep them.

8. Saving and resuming progress

Cascade keeps your work in your current session. To carry on another day, on another device or after a break, save a progress file and import it later. Nothing needs to be uploaded or cleaned a second time.

Save Progress

1. Open the Visualize page.
2. Press **Save Progress**. It is in the top row of buttons and at the bottom of the Contents list on the left.
3. Your browser downloads a file named like `cascade-progress_sales_2026-10-18.cascade`. Keep it somewhere safe.

Save Progress is only on the Visualize page, so finish cleaning first.

Saved in the file	Not saved
The cleaned data (after your cleaning choices)	The original file you uploaded
All charts and their settings	Anything you typed but did not press Save on
Chart insights, pivot notes, Written Summary and Recommendations that you edited	Automatic text. Cascade writes it again when you open the file.
Pivot tables, filters, Key Metrics and the page order	Light or dark mode (your browser remembers it)
Currency symbols and the cleaning record shown in File Overview	The PDF. Download it again from the restored page.

Import Progress

There are two ways in, and both end up on the Visualize page.

From the Visualize page

1. Press **Import Progress** in the top row.
2. Choose your .cascade file.
3. Cascade asks "Import …? This replaces everything on this page: the data, charts, notes and pivot tables." Press OK only if you have saved what you are working on.

From the Upload page

1. Go to **Continue saved work**.
2. Press the file box (**Select a Cascade progress file (.cascade)**) and choose your file.
3. Press **Import & Continue**.

When it works you see a green notice such as "Restored your saved progress from 2026-10-18: 3 charts, 1 pivot table and 1,200 rows of data." Press × to dismiss it.

If an import fails

Cascade checks the whole file first and changes nothing unless it passes, so a failed import never damages your current work.

Message (shortened)	What to do
Over the 15 MB limit for progress files	The file is too big. Save a new one from a smaller dataset.
Not a Cascade progress file	Use a file made by Save Progress that still ends in .cascade.
Damaged or changed after it was saved	Do not edit or re-zip the file. Save it again from Visualize.
Saved by a newer version of Cascade	Open it in the newer version, or save it again there.
Contains no data, or has too many columns	The file is empty or beyond Cascade's limit of 500 columns.

Keep in mind

- Press each box's own Save button before you press Save Progress. Text marked Custom (unsaved) is not included.
- The file is a snapshot. Save again after you make changes. A new save is a new file, so your browser may add (1) to the name.
- The file holds your whole cleaned dataset and your notes. Treat it with the same care as the original data and do not share it with people who should not see it.
- Do not open, rename the ending of or edit the file. Cascade needs it exactly as it was saved.
- Opening a file brings back the data and settings, not a separate copy of the original upload. Your cleaning choices are shown in File Overview.
- Settings that no longer fit the data are skipped on import instead of stopping it. Very long notes are shortened (5,000 characters for a chart or pivot note, 20,000 for the Written Summary, and 100 recommendation lines).
- Cascade restores up to 50 charts, 20 pivot tables, 12 filters and 12 Key Metrics.

9. Exporting the PDF report

The PDF is Cascade's finished product: a cover, a contents page and one section for each chart and pivot table, ready to hand to someone else.

How to export

1. Check the Visualize page first. The PDF shows what the page shows, so settle your filters, charts, pivots and notes.
2. Press **Download PDF**. It is in the top row of buttons and at the bottom of the Contents list.
3. Cascade opens the report page and your browser's print window appears by itself after a moment.
4. Set the destination to **Save as PDF** (in some browsers, "Microsoft Print to PDF") and press **Save**. Your browser suggests a file name starting with "CASCADE Report -". You can change it.
5. To go back, press **Back to Visualizer** at the top of the report page. If the print window did not open, press **Save as PDF** on that page.

If colours or the cover art look missing, turn on **Background graphics** in the print window. If a web address or date shows at the top or bottom of the pages, turn off **Headers and footers**. The pages are A4 portrait.

What the report contains

Part	What is in it
Cover	The title "Business Insights & Analysis", the file name, your name (full name, or your username if none is set) and today's date

Part	What is in it
Contents	A numbered list: Overall Data Summary & Key Metrics, every chart and pivot table in page order, then Recommendations
01. Overall Data Summary & Key Metrics	Dataset name, record count, column count, number of charts, a Data Preparation note about what was done to missing values, your Key Metrics as cards and the Summary text
Each chart	"Visual #N" and its title, the chart picture, Key Insights & Findings, and any recommendations written for that chart
Each pivot table	The table with a Rank column, the totals row and the notes under it, then Key Insight and Recommendation boxes
Final section: Recommendations	A numbered list of all recommendations, each labelled with its chart, Data Quality or General

Keep in mind

- The PDF is built from your current page every time. Change something, then export again.
- It uses your filtered rows, and the active filters are not printed on the report. If your reader needs to know about them, say so in the Written Summary.
- The File Overview preview, the cleaning report and the Filters panel are not in the PDF. Only a short Data Preparation sentence is.
- A pivot table that shows an error on the page is left out of the PDF. A filter that leaves no rows sends you back to the Visualize page instead of making a report.
- Charts are the same pictures as on the page. A chart with a calculation, grouping or Top N is printed as you see it.
- Wide pivot tables are printed in smaller type so they fit the page. If one is hard to read, remove some columns.
- Before you send the PDF, scroll the Visualize page once to read every insight, since the PDF prints them as written.
- Save Progress does not save the PDF. Export the PDF, then press **Save Progress** if you want to be able to rebuild it later.

10. Limits and what Cascade can and cannot do

Cascade is a data preparation and exploratory reporting tool. It helps organize tabular data, calculate summaries, build charts and pivot tables, and prepare a report. It does not replace checking the data or using subject-matter judgment.

Capability and limitation table

Cascade can…	Cascade cannot… / important limit
Import CSV, Excel (.xlsx/.xls), JSON, JSON Lines (.jsonl) and NDJSON (.ndjson) files.	Import every file format. PDFs, Word documents, images and database connections are not supported as dataset uploads.
Read a single tabular dataset.	Combine multiple files or Excel sheets automatically. For Excel, the first sheet is read.
Clean blank rows, exact duplicates, spacing, common missing-value markers, many numeric-looking columns, column names and certain date formats.	Know whether a value is factually correct, whether two identical records are genuinely duplicates, or whether a missing value should be filled with a particular value. Review the cleaning report.
Flag possible numeric outliers and near-duplicate rows.	Decide automatically that an outlier is an error. Near-duplicate checks are only performed for datasets with up to 1,000 rows.

Cascade can…	Cascade cannot… / important limit
Let you remove rows with missing values or review and edit them manually.	Manually review more than 500 rows with missing values in one cleaning session. For larger cases, remove those rows or fix the source file and upload it again.
Create bar, horizontal bar, pie, line, area, scatter, histogram, box and correlation heatmap charts.	Make every chart type suitable for every dataset. Some chart types require numeric, date or category columns; an unsuitable combination may produce a warning or a different chart.
Group date-based line, area and bar charts by day, week, month, quarter or year.	Automatically know the business meaning of a date, currency, quantity or metric. Check that the selected aggregation matches your question.
Calculate a value from two columns or a number using addition, subtraction, multiplication or division.	Calculate two separate custom expressions at once. Only one calculation can be active per chart. Division by zero is skipped. Pie charts cannot meaningfully display calculated ratios or averages, so Cascade switches that result to a bar chart.
Build pivot tables using one row field, an optional column field and one measure.	Build multi-level pivot hierarchies. With a column field, only the 15 columns with the largest totals are displayed; Show all displays at most 200 rows.
Filter the dataset and apply the filters to metrics, charts, pivots and the PDF.	Keep separate filters for the same column. Adding a filter for a column replaces the existing filter for that column. Up to 12 filters are kept.
Create and edit Key Metrics, summaries, chart insights and recommendations.	Guarantee that automatic insights are a complete business explanation or a substitute for checking the figures. Review generated text before sharing it.
Save a working snapshot in a .cascade progress file and restore it later.	Save unsaved text, the original upload, theme preference or the PDF inside that progress file. Press each section's Save button before saving progress.
Export a PDF through the browser's print dialog and download pivot tables as CSV.	Guarantee that the browser will automatically open the print dialog in every browser configuration. Use the report page's **Save as PDF** button if needed.

Hard limits at a glance

Item	Limit / behavior
New data upload	5 MB maximum
Saved progress file	15 MB maximum
Data stored inside a progress file	50 MB maximum when unpacked
Columns in a restored progress file	500 maximum
Charts restored from progress	Up to 50
Pivot tables restored from progress	Up to 20
Filters	Up to 12
Key Metrics cards	Up to 12
Top N for charts and pivots	1 to 50
Pivot table display	Up to 200 rows; up to 15 displayed pivot columns when a Columns field is used
Manual missing-value review	Up to 500 affected rows

Item	Limit / behavior
Near-duplicate check	Only for datasets of up to 1,000 rows
Saved chart or pivot note	Up to 5,000 characters
Saved Written Summary	Up to 20,000 characters
Saved recommendations	Up to 100 lines

These limits describe the current version of Cascade represented by this manual. If the application is updated, the exact limits or wording may change.

11. Troubleshooting: common messages, causes and fixes

Start with the message shown on the page. If the page freezes or displays a generic server error instead of a helpful message, note the action you took and the file type/size, then try the safe checks below before repeating the action.

Upload and import messages

Message / symptom	Likely cause	Fix
Please choose a file to upload.	No file was selected.	Choose a file in the upload box, then submit again.
That file is empty. Please choose a file that contains data.	The selected file has no content.	Open it in the source application and export or save the correct dataset.
Over the 5 MB limit	The uploaded file exceeds the upload cap.	Remove unused rows or columns, export only the needed data, or split it into smaller files. Remember that Cascade processes one dataset at a time.
Unsupported file type.	The file extension is not supported.	Export the data as .csv, .xlsx, .xls, .json, .jsonl or .ndjson.
That file is not valid JSON / Could not turn that JSON into a table	The JSON is malformed or does not contain a supported tabular structure.	Use a list of similarly structured records, for example [{"region": "North", "sales": 100}, {"region": "South", "sales": 150}]. For JSON Lines, put one JSON record on each line. Check commas, quotation marks and brackets.
Could not read that file. Make sure it is valid JSON.	The JSON parser could not read the file.	Re-export it as valid JSON or try CSV if the data is a simple table.
The uploaded Excel data looks incomplete or unexpected.	The workbook has multiple sheets, title rows, merged cells or totals around the data.	Move the actual table to the first sheet, use one header row, unmerge cells and remove decorative title/total rows before uploading.
Values or characters look incorrect in a CSV.	The file may use an unexpected text encoding or delimiter.	Re-export as a comma-separated UTF-8 CSV. If the file still imports incorrectly, open and resave it using a spreadsheet application.

Cleaning messages and unexpected results

Message / symptom	Likely cause	Fix
Manual review is limited to 500 rows at a time.	More than 500 rows contain missing values.	Choose the remove-rows option, or correct missing values in the source file and upload it again.
Choose Edit or Remove for this row.	At least one affected row has no decision.	Return to the row and select **Edit** or **Remove row**. Every affected row needs a decision.
A manual value is rejected.	The value does not match the detected column type, or it is still treated as missing.	Use a whole number/number for numeric columns, YYYY-MM-DD for a date, YYYY-MM for a month, or valid text. Do not enter NA, null or - as a replacement for a missing value.
A column name changes after cleaning.	Cascade normalizes column names.	Use the cleaned name shown in File Overview, usually lowercase with spaces replaced by underscores, when setting up charts and pivots.
Some rows disappeared.	Empty rows, exact duplicates or rows with missing values may have been removed after confirmation.	Read the Cleaning Report and compare its row counts and missing-value decisions. If legitimate records were identical in every column, add a unique record ID to the source and upload again.
A large number is flagged as an outlier.	The value is far from the rest of the numeric values.	Check the original record and its context. Keep valid exceptional values; correct or remove a value only when you have evidence it is wrong.

Charts, filters and pivot tables

Message / symptom	Likely cause	Fix
No data matches the current filters or charts are empty.	The active filters exclude every row, or the selected fields contain no usable values.	Remove one or more filters or press **Clear all**. Confirm the row count and check the selected columns.
No valid numeric data was found… / **No valid numeric values were found…**	The selected measure has no usable numeric values after cleaning, filtering or calculation.	Select a numeric column, adjust the filters, or check the source values. For calculated values, verify both operands and avoid division by zero.
No valid data was found… / **No categorical values were found…**	The selected X/Y fields have no usable values for the requested chart.	Choose different columns or clear filters. A category-versus-number chart generally needs a category column and a numeric measure.
The chart type changes from Pie to Bar.	Pie charts cannot display the selected calculated ratios or averages appropriately.	Use the resulting bar chart, or choose a measure that represents category values/shares suitable for a pie chart.
The chart shows only some categories.	The chart has a display cap or a Top N / Bottom N setting.	Check the note on the chart and the Sort & limit controls. Use Top N up to 50, or select a chart type better suited to the number of categories.

Message / symptom	Likely cause	Fix
Dates appear out of order or the trend looks wrong.	Dates may be stored as text, may not be recognized, or the chart is showing raw rows rather than grouped periods.	Check the cleaned date column. For a date-based Auto, Line, Area or Bar chart, choose a date column for X and set **Group dates by** to the appropriate period.
A pivot table omits categories or rows.	Pivot display limits apply.	Use Top N / Bottom N, add a filter, group dates into a larger period, or remove the optional Columns field to simplify the table. With a Columns field, only 15 columns are displayed; Show all is capped at 200 rows.
An insight or summary no longer contains your custom text.	A chart, filter, Key Metric or pivot setting changed, causing related text to be regenerated.	Recheck the current figures, then rewrite and save the text. Use the reset behavior table in section 7 before changing settings.
A pivot table displays an error or is missing from the PDF.	The pivot may not be valid for the current data/settings; pivots with errors are omitted from PDF output.	Reopen the pivot settings, choose valid columns and a supported calculation, press **Update Pivot**, then retry the export.

Save Progress and PDF messages

Message / symptom	Likely cause	Fix
Over the 15 MB limit for progress files	The saved file is too large.	Save progress from a smaller dataset, if possible. Do not manually edit or repackage the .cascade file.
Not a Cascade progress file	The selected file is not a valid Cascade snapshot.	Select a file downloaded using **Save Progress**. Keep the .cascade extension unchanged.
Damaged or changed after it was saved / progress file could not be opened	The file may be incomplete, altered or damaged.	Try the original download. If unavailable, reopen the source work and create a new progress file. Do not rename its extension or re-zip it.
Saved by a newer version of Cascade	The file was created in a newer app version.	Import it using that newer version of Cascade, if available, then save a new snapshot there.
The data inside that progress file has too many columns	The restored dataset has more than 500 columns.	Reduce the source to the columns needed and create a new progress file.
Custom text does not appear after importing progress.	The text was not saved before saving progress, or the progress file is an older snapshot.	Return to the original session if possible, press the relevant Save button, then press **Save Progress** again.
The PDF print window does not open.	The browser blocked or failed to open the print dialog.	On the report page, press **Save as PDF**. Allow pop-ups for Cascade if your browser asks.
PDF colours or cover artwork are missing.	Background graphics are disabled in the print settings.	Turn on **Background graphics** in the browser's print dialog.
A URL or date appears in the PDF header/footer.	Browser print headers and footers are enabled.	Turn off **Headers and footers** in the print dialog.

Message / symptom	Likely cause	Fix
The PDF has smaller text in a pivot table.	The table is wide and has been fitted to the page.	Remove unnecessary pivot columns, use a simpler pivot or download the pivot as CSV for closer inspection.

If the problem continues

1. Avoid repeatedly clicking a button if the page appears to be processing.
2. Save Progress if Visualize is still working and the page is responsive.
3. Record the exact message, what you clicked immediately before it appeared, the file type and approximate file size. Do not include private or confidential data in a support message.
4. Try a fresh export from the original source file if the file itself may be malformed.
5. If the page shows a generic server error or the same operation repeatedly fails, report the steps to the person maintaining the Cascade deployment. A generic server error may require checking the server logs; the manual cannot diagnose it from the browser message alone.

12. Good habits for reliable results

Before uploading

- **Keep a clean source copy.** Make changes in a duplicate of the original file, not the only copy.
- **Use one clear header row.** Give every column a meaningful name and avoid merged cells, decorative titles and subtotals inside the dataset.
- **Use one record per row.** Do not mix totals, notes or blank separators into the data table.
- **Include an ID when records matter individually.** Two rows identical across every column are treated as duplicates, even if they represent separate real-world events.
- **Check types and units.** Do not mix text and numbers in the same measure column if you expect to calculate totals or averages. Use consistent date formats and units.
- **Remove unnecessary columns and rows.** This helps keep the upload under 5 MB and makes charts easier to configure.
- **Protect sensitive data.** Files are stored on the server running Cascade. Upload only information you are authorized to store there.

While cleaning and analyzing

- **Read the Cleaning Report before moving on.** Check how many rows were removed, how column names changed and how missing values were handled.
- **Do not remove data automatically just because it looks unusual.** Verify outliers and duplicate-like rows against the source and your knowledge of the data.
- **Finish filters before writing interpretations.** Filters affect Key Metrics, every chart, every pivot table and the PDF.
- **Choose a measure that answers the question.** For example, sum revenue to compare total revenue, count records to compare order volume, and average rating to compare typical ratings. A sum of unit prices may not be meaningful.
- **Choose a chart for the question.** Use bars for category comparisons, lines for trends over time, scatter plots for relationships between numeric measures, histograms for distributions and heatmaps to inspect numeric correlations.
- **Check aggregation settings.** Daily/monthly grouping with Sum answers a different question from Average, Count, Maximum or Minimum.
- **Look at the row count after filtering.** A chart that looks empty may be correctly reflecting a filter that removed all matching records.
- **Treat automatic insights as a draft.** Confirm the chart, metric, filters, units and time period before using the wording in a presentation or decision.
- **Avoid overcrowding.** Use Top N, filters or a coarser date grouping instead of forcing too many categories into one chart.

Before saving or sharing

- **Save custom text explicitly.** A label such as **Custom (unsaved)** means your changes are not stored yet.
- **Save Progress after meaningful milestones.** It is a snapshot; make a new one after important changes. Store it securely because it contains the cleaned dataset and notes.

- **Check the PDF end to end.** Verify the title, figures, chart labels, pivot totals, written insights and recommendations. Remember that active filters are not listed in the PDF, so mention important filter conditions in your Written Summary when the audience needs them.
- **Export again after changes.** The PDF is generated from the current page and is not automatically updated when you change the analysis.
- **Keep a reproducible record.** Retain the original dataset, the Cleaning Report details, the `.cascade` progress file and the final PDF when appropriate.
- **Share only what the audience needs.** Check the report and progress file for confidential information before sending them to anyone else.

Final review checklist

- ☐ I checked the row and column counts in File Overview.
- ☐ I reviewed the Cleaning Report and missing-value decisions.
- ☐ I confirmed that filters match the question I am answering.
- ☐ I checked that each Key Metric and pivot calculation is appropriate.
- ☐ I reviewed every chart, including labels, units, date grouping and category limits.
- ☐ I verified the automatic insights and edited any wording that could mislead the reader.
- ☐ I saved every custom text box before saving progress.
- ☐ I exported and reviewed the final PDF.
- ☐ I checked that the files I am sharing do not expose data unnecessarily.